

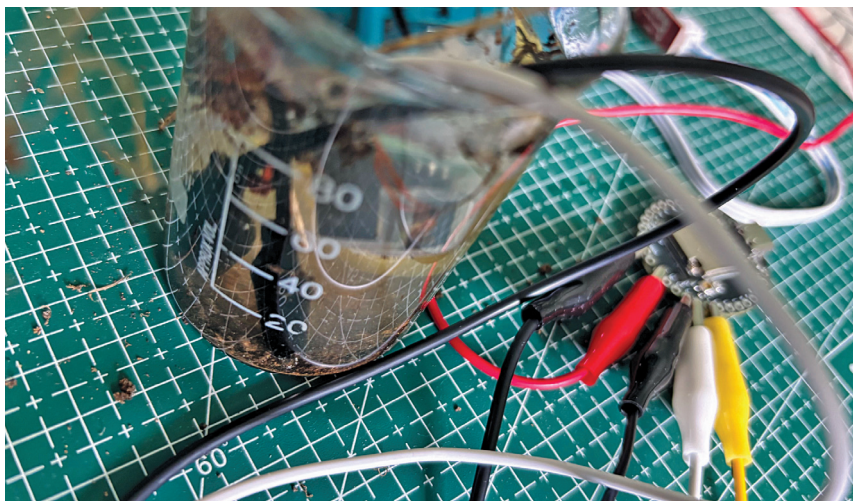


Fig. 4: Installation of the sensor on the tanks

vertically within the tank and connected to pins 3, 4, and 5. Probes must be clean, insulated, and not in physical contact with one another. The TDS probe should be immersed approximately halfway, while the turbidity sensor must remain aligned and submerged during measurement.

Power is supplied via a 5V adaptor. After uploading the program, the serial monitor or web dashboard can be used to observe live data. Gradually filling the tank will allow level readings to update. Introducing salt increases TDS readings, while adding particles

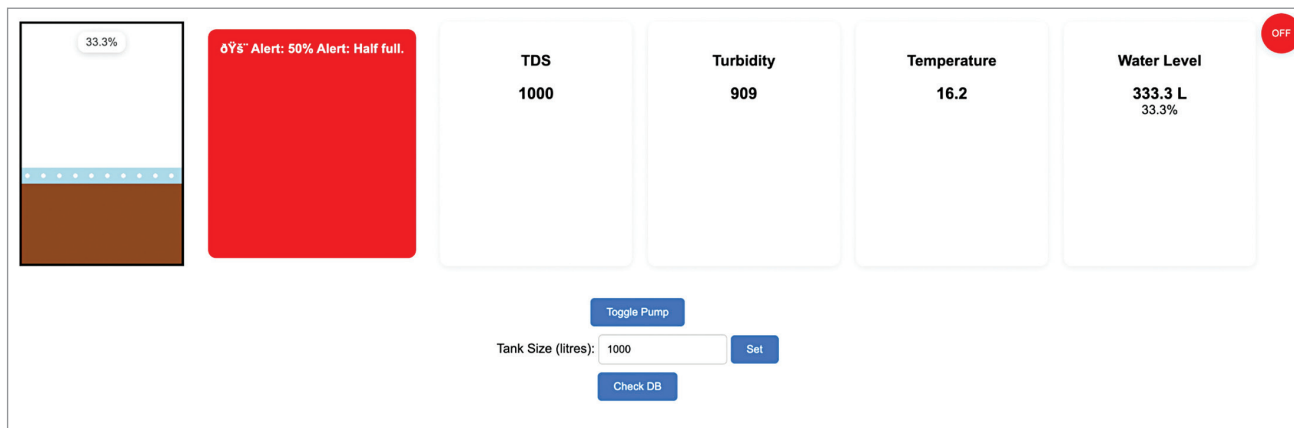


Fig. 5: Dashboard showing the water level data

Sensor pins are assigned within the code, and threshold values are defined for turbidity, TDS, temperature, and water level. The web dashboard and database functions are preconfigured to process and display readings in real time. After final verification, the code is uploaded through the Arduino IDE by selecting the appropriate board and communication port.

Finally, after configuring the code, it is uploaded to the IndusBoard Coin by selecting the correct board and port in the Arduino IDE settings.

Construction and testing

The IndusBoard Coin V2, TDS sensor, turbidity sensor, and relay module may be mounted on an acrylic or wooden base. Stainless steel probes should be positioned

such as dust or soap increases turbidity. When configured thresholds are reached, the pump is automatically switched on or off. Once stable operation is confirmed, the system may be enclosed in a 3D-printed or moulded case for long-term installation. **EFY**

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